Resource Hosting & Datacenter

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Outline

Resource Hosting

Main Components in a Datacenter

Datacenter Configuration

Resource Hosting

On-Premise

- Organisations are responsible for deploying, operating and maintaining internet connectivity
- Have complete control over Quality of Service (QoS)
- Providing security

Cloud-Based

- Use multiple cloud carriers – QoS dependent on carriers
- Easier to adopt for applications with relaxed bandwidth and latency
- Security concerns

not inherently insecure
just more actors

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Terminologies

Latency

Time it takes a packet to travel from one data node to another – important when response time is key for applications

Bandwidth

Number of bits transferred per unit time – important when a substantial amount of data transfer is required by the application

`easier to increase bandwidth than latency (physical property)`

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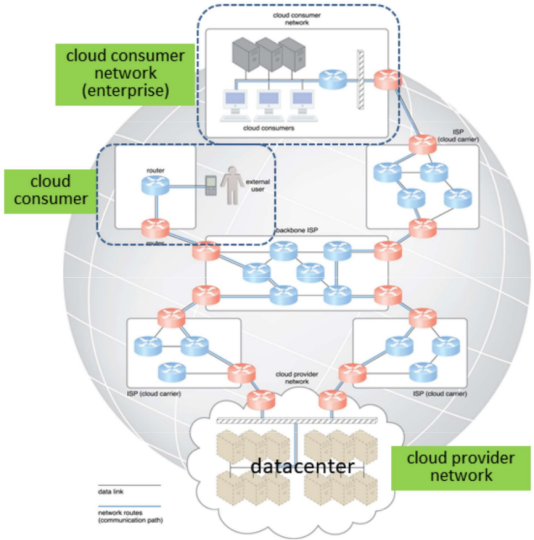
On-Premise vs Cloud-Based

Time Scales of Computing Latencies

Event	Latency	≈ Scaled Time
1 CPU cycle	0.3 ns	1 second
Level 1 cache access	0.9 ns	3 seconds
Main memory access (DRAM, from CPU)	120 ns	6 minutes
Solid-state disk I/O (Flash memory)	50 – 150 μs	2 – 6 days
Rotational disk I/O	1 – 10 ms	1 – 12 months
Internet: Singapore to Hong Kong (2555 Km)	40 ms	4 years
Internet: Singapore to London (10856 Km)	155 ms	16 years
Flight: Singapore to Tokyo (5322 Km)	7 hrs	2.5 million years!

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Cloud Connectivity



importance of cloud carriers or ISPs

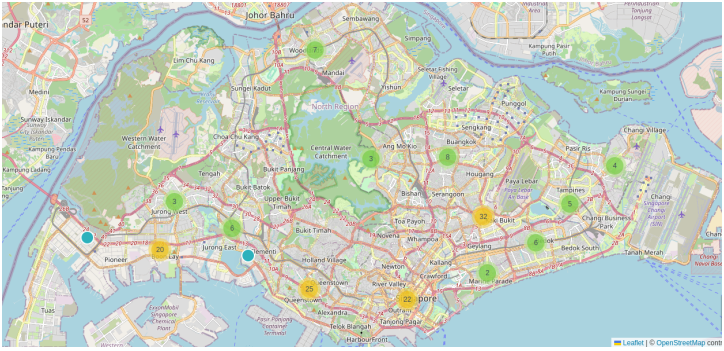
CDN:
have versions of the content closer to the consumers

Singapore Data Centers Locations

How Many?

145 Locations

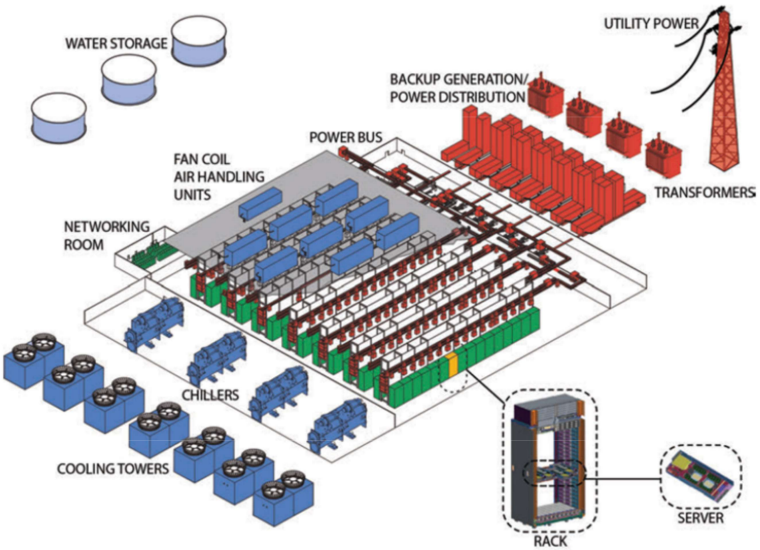
50 Providers



https://www.datacenters.com/locations/singapore

Datacenter I

Components

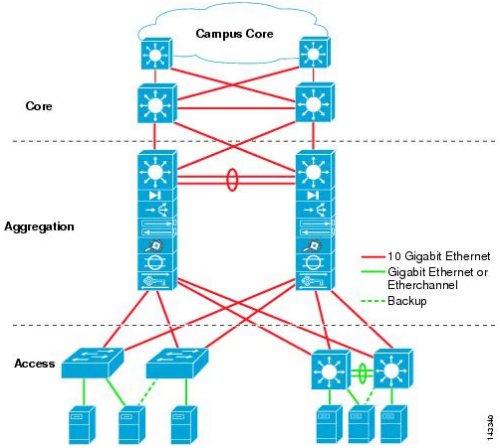


Main server hall – compute, storage, and networking equipment

Mechanical yard – hosts cooling systems

Electrical yard – generators and power distribution centers

Datacenter Servers I



organization: layering

https://www.cisco.com/c/en/us/td/docs/solutions/Enterprise/Data_Center/DC_Infra2_5/DCI_SRND_2_5a_book/DCInfra_1a.html

Datacenter Servers II

like lego -> building blocks

Uses commodity hardware with a modular architecture

Servers are modularized and packed as a cluster in server racks

Server racks are replicated as scaled up computational resources to form a datacenter

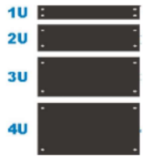
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Datacenter Servers III

a. server

1U Server (1.75 inches or 44.45mm)





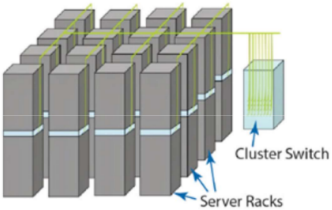
1 rack unit
Standardized amount of space

b. cluster



42U Server Rack

c. Multiple Clusters



d. datacenter



Datacenter Servers

Example – Supermicro AS -1127H7-N — 1U Hyper Server

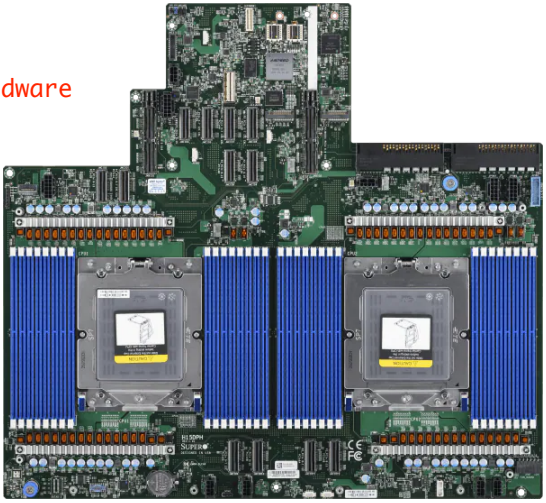


https://www.supermicro.com/en/products/system/hyper/1u/as-1127h7-n

Datacenter Servers

Example – Supermicro AS -1127H7-N — 1U Hyper Server

Actually generic hardware
Not specialized



2 CPUs
Connected by buses

Blue -> memory slots

https://www.supermicro.com/en/products/system/hyper/1u/as-1127h7-n

Dual processor(s) AMD EPYC™ 9006 SP7 Series Processors

- Up to 512C/1024T 512 cores/ 1024 threads

32 DIMM slots memory

3 PCIe expansion slots for networking and storage

Flexible Networking via AIOM (Advanced I/O Module)

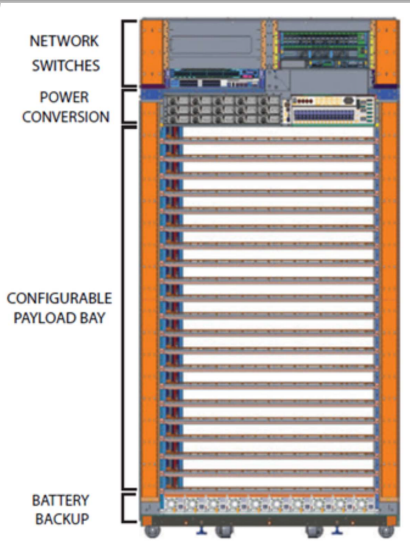
...

Supports servers, storage and networking equipment

Has provision for power, battery backup and rack-level switch (e.g., Top Of Rack – TOR)

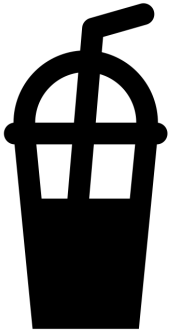
Rack-level switch – connects IT resources within rack and also connects the rack to the datacenter network

Typical commodity rack can house 42 1U servers, each of 1.75in height



rack -> connections
reduce latency -> keep workloads local

Take a Break







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Two **categories**

Private to **individual running tasks** – resides in local DRAM or disk

Part of shared state of **distributed workload** – distributed storage network

Storage Hierarchy

One server – caches, DRAM memory, Disks, Flash Disks

Each rack – local and distributed memory

Across racks – local and distributed memory

Datacenter Network

Carrier Interconnection

Between datacenter (LAN) and cloud consumers (WAN)

Local Area Network (LAN)

Within a rack

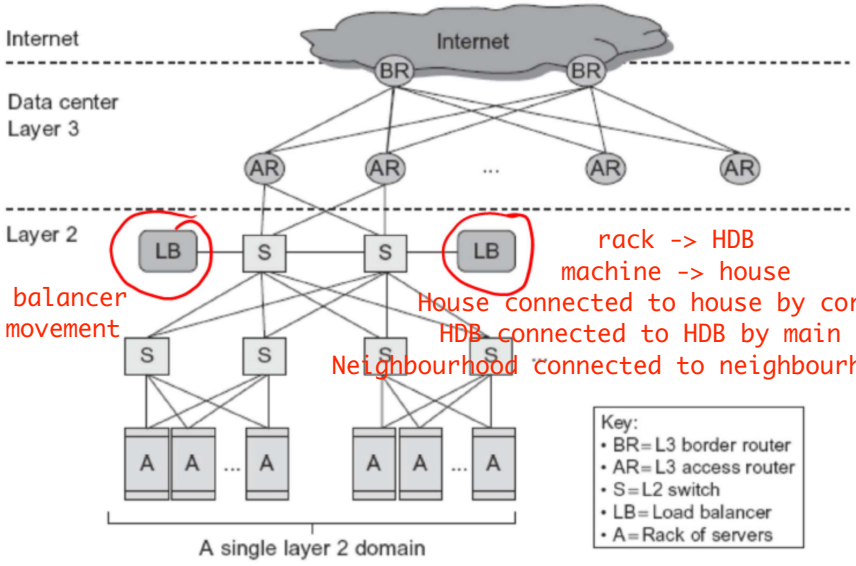
Redundant connectivity

Storage Area Network (SAN)

Between servers and storage systems

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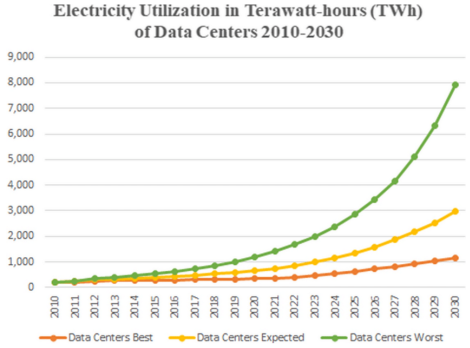
Datacenter Network



network load balancer
manage data movement

rack -> HDB
machine -> house
House connected to house by corridor
HDB connected to HDB by main road
Neighbourhood connected to neighbourhood by road

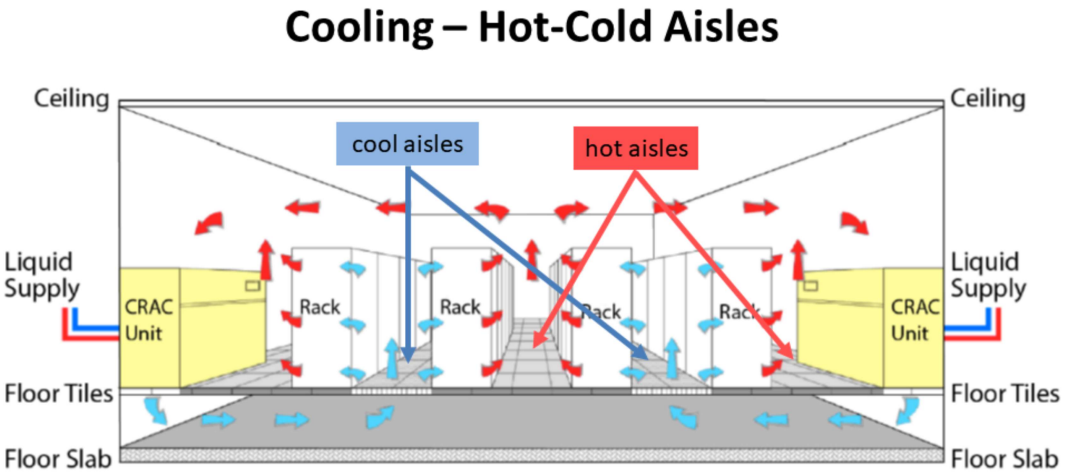
Datacenter Electricity Demand



https://link.springer.com/article/10.1007/s10586-021-03431-z

Datacenters consume a **lot** of energy

Most of this is converted into heat – cooling is a big issue!



Cooling system in a raised-floor datacenter with hot-cold air circulation support water heat exchange facilities

Datacenter Cooling

From air to liquid (water) cooling – Water is 1000x more efficient than air to remove heat

From portable water to seawater

- Planned Singapore Floating Datacenter Park (`https://www.keppeldatacentres.com/innovations/floating-data-centre/`)

In the sea?

- Absolutely! (Microsoft Project Natick was the first Proof of Concept – `https://www.microsoft.com/en-us/research/project/natick/`)
- The 'Cloud' in the Ocean ☺

Water Usage!

Very high –
`https://www.eesi.org/articles/view/data-centers-and-water-consumption`

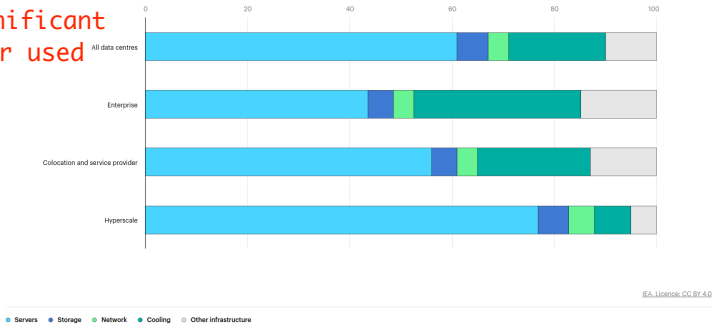
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data center in space? possible

Datacenter

Power Usage Efficiency

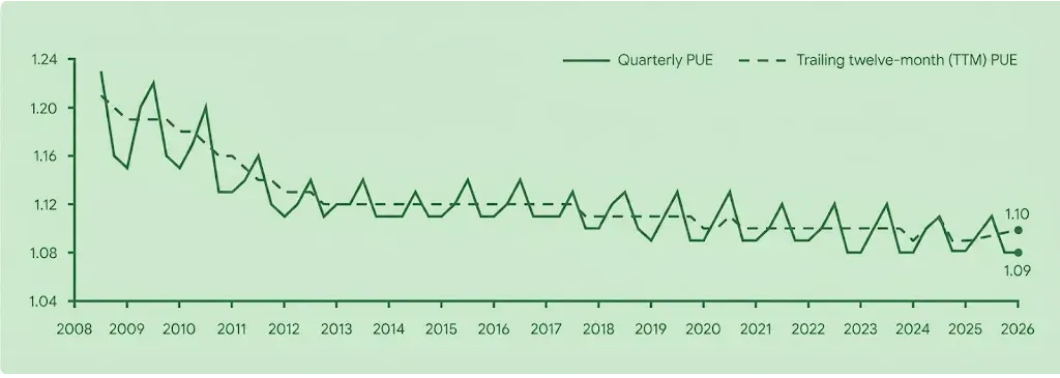
cooling is a significant portion of power used



https://www.iea.org/data-and-statistics/charts/share-of-electricity-consumption-by-data-centre-and-equipment-type-2024

$$PUE = \frac{total\ power\ used\ by\ datacenter}{power\ used\ by\ IT\ Equipment} \Rightarrow \text{in ideal case PUE is 1}$$

Average PUE for all data centers



https://datacenters.google/efficiency/

Energy Proportional Systems

Energy Efficiency is not a linear function of load – idle system uses 50% power

An energy proportional system consumes almost no power when idle, very little power under light load, and, gradually, more power as the load increases

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Provides a ranking system to indicate the reliability of a datacenter

Tier	Description	Expected Uptime
1	Single path for power distribution, USP, and cooling distribution, no redundant components	99.671%
2	Partial redundancy for power and cooling	99.741%
3	Multiple paths for power and cooling, and each path with redundant components, concurrently maintainable	99.982%
4	Redundancy for every component	99.995%

Availability & Cost

Tier 1 < Tier 2 < Tier 3 < Tier 4

Tier 3 is typically industry standard
Tier 4 might be overkill

SoC Datacenter Visit

22/09/26 from **2pm**

24/09/26 from **7pm**

Register your interest: `https://forms.cloud.microsoft/r/dC2vmB0f3W`
(available from: **01/09/26 at 20:00 – 08/09/26 at 20:30**)



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Next Week

Virtualisation and Multitenancy

References

Chapter 5 and Appendix D1, Cloud Computing: Concepts, Technology & Architecture, Thomas Erl, Zaigham Mahmood and Ricardo Puttini, Prentice-Hall, 2013

Chapters 1 & 4, The Datacenter as a Computer: Designing Warehouse-Scale Machines, 3rd Edition, Morgan & Claypool Publishers, 2019

Continuous Feedback



https://forms.cloud.microsoft/r/ZJxTf2wFjr

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